Ozgur Ural, Ph.D.

Curriculum Vitae, November 1, 2026

Leiden, Netherlands • drozgurural@gmail.com • +31 6 2903 0434

ozgurural.github.io • Google Scholar • ORCID 0000-0003-1329-4303 • linkedin.com/in/uralozgur

Machine learning researcher working on **trustworthy AI for mission-critical systems**: evidence of how a model was trained and who owns it, robustness against attackers, and predictable timing when a model runs inside a real-time loop. Ph.D. in Electrical Engineering and Computer Science, Embry-Riddle Aeronautical University (2025), with a machine-learning dissertation on Proof-of-Learning and model watermarking and three first-author articles in IEEE Access. Over twelve years of industry experience building production systems in cybersecurity, defense and flight simulation, currently at Avion Full Flight Simulators, leading the company's enterprise AI strategy.

RESEARCH INTERESTS

- Trustworthy and verifiable machine learning: Proof-of-Learning, model watermarking, model provenance and ownership verification
- Security of machine learning: spoofing attacks on training records, adversarial robustness
- Blockchain-enhanced machine learning
- Machine learning in real-time and safety-critical systems
- Natural language processing for cyber security

EDUCATION

Ph.D. in Electrical Engineering and Computer Science Aug 2021 – Aug 2025

Embry-Riddle Aeronautical University, Daytona Beach, Florida, USA

- Dissertation: Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking.
- Advisor: Dr. Kenji Yoshigoe (Committee Chair); committee included IEEE Fellow Dr. Houbing Song.

M.Sc. in Cyber Security 2015 – 2019

Middle East Technical University (METU), Ankara, Turkey

- Thesis: Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Turkish Newspaper Data.
- Advisor: Prof. Cengiz Acartürk.

B.Sc. in Computer Engineering 2010 – 2014

Middle East Technical University (METU), Ankara, Turkey

- English-medium programme; entered among roughly the top 1,000 of more than 2 million national candidates.

Bachelor of Business Administration 2012 – 2017

Anadolu University, Eskisehir, Turkey

PUBLICATIONS

Google Scholar (November 2026): 51 citations, h-index 4.

Journal Articles

[J3] **Ural, O.** and Yoshigoe, K. (2025). SecurePoL: Integration of Watermarking With Proof-of-Learning to Enhance Security Against Spoofing Attacks. *IEEE Access*, vol. 13, pp. 213067–213091. DOI: 10.1109/ACCESS.2025.3642198. *Research article.*

[J2] **Ural, O.** and Yoshigoe, K. (2024). Enhancing Security of Proof-of-Learning Against Spoofing Attacks Using Feature-Based Model Watermarking. *IEEE Access*, vol. 12, pp. 169567–169591. DOI: 10.1109/ACCESS.2024.3489776. *Research article.*

[J1] **Ural, O.** and Yoshigoe, K. (2023). Survey on Blockchain-Enhanced Machine Learning. *IEEE Access*, vol. 11, pp. 145331–145362. DOI: 10.1109/ACCESS.2023.3344669. *Survey article.*

IEEE Access: 2025 Journal Impact Factor 4.2 (JCR 2026).

Conference Papers

[C2] **Ural, O.** and Acartürk, C. (2021). Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. In *Proceedings of the 7th International Conference on Information Systems Security and Privacy (ICISSP 2021)*, pp. 66–76. DOI: 10.5220/0010201600660076.

[C1] Yamaç, A. Ö., Doçi, J., **Ural, O.**, Özarslan, Ö. F. and Temizer, S. (2014). Clover: Autonomous Cargo and Mail Delivery. *Turkish Autonomous Robots Conference (TORK 2014)*, Ankara, Turkey.

Theses

[T2] **Ural, O.** (2025). Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Doctoral dissertation, Embry-Riddle Aeronautical University.

[T1] **Ural, O.** (2019). Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Turkish Newspaper Data. Master's thesis, Middle East Technical University.

Technical Reports (not peer reviewed)

[R1] **Ural, O.** and Erdur, E. (2016). Secure Proxy on Cloud. Technical report. DOI: 10.13140/RG.2.2.24058.08649.

Narrated explainers of this research: ozgurural.github.io/lab.

RESEARCH EXPERIENCE

Graduate Research Assistant Sep 2021 – Dec 2024, May 2025 – Oct 2025

Embry-Riddle Aeronautical University, Daytona Beach, Florida, USA. Advisor: Dr. Kenji Yoshigoe

- Original research on verifiable, trustworthy machine learning across Proof-of-Learning, model watermarking and adversarial robustness, published as three first-author IEEE Access articles [J1–J3] and the doctoral dissertation [T2].

M.Sc. Thesis Research 2015 – 2019

Middle East Technical University, Ankara, Turkey. Advisor: Prof. Cengiz Acartürk

- Automatic detection of cyber security events from Turkish Twitter and newspaper data [T1], published at ICISSP 2021 [C2].

PROFESSIONAL SERVICE

- **Program Committee (2026):** NLPAICS 2026, 2nd Workshop on NLP Applied to Information and Cyber Security, Alicante.
- **Reviewer:** IEEE Access and other journals; IEEE and ACM conferences (security, machine learning, distributed systems).

AWARDS

- First Prize, TUBITAK national project competition (2014).
- Second Prize, METU Computer Engineering graduation Demo Day (2014).

INDUSTRY EXPERIENCE

Senior Software Engineer Oct 2023 – Present

Avion Full Flight Simulators, Nieuw-Vennep, Netherlands

- Lead Avion's enterprise AI strategy: authored the AI adoption roadmap and a three-tier architecture spanning air-gapped on-prem LLMs, a governed cloud-LLM API tier, and edge deployment.
- Designed the strategy's proposed AI workflows, each keeping a person accountable for the outcome; built engineering-knowledge retrieval prototypes and the strategy portal end to end.
- Developed real-time simulator software in C, C++ and Scala, including a low-latency telemetry pipeline sustaining approximately 50 GB/s with batching and bounded-queue backpressure.

Software Team Lead Nov 2020 – Aug 2021

Havelsan, Ankara, Turkey

- Led a team of 14 engineers (developers, QA, DevOps, support) delivering the Havelsan DLP endpoint Data Leakage Prevention product; developed core endpoint-agent features in C, C++, Java and Go.

Expert Software Engineer Feb 2019 – Nov 2020

STM Defence Technologies, Ankara, Turkey

- Developed mission-control and ground-control components for the Kargu and Togan autonomous UAVs in C++11 with Boost and Qt, under hard-real-time, safety-critical constraints; designed Power-On and Continuous Built-In Tests.

Expert Software Engineer Jul 2014 – Feb 2019

Comodo Cybersecurity, Ankara, Turkey

- Led design and development of Windows services and applications in C++17 with Boost and WinAPI (Comodo Dome Secure Web Gateway, Comodo Patch Manager, Comodo Dragon browser); established unit testing, code review and Jenkins CI/CD.

TECHNICAL SKILLS

- **Machine learning:** PyTorch, TensorFlow, scikit-learn, deep learning, LLMs, retrieval-augmented generation, federated learning; model watermarking, Proof-of-Learning, adversarial robustness
- **Languages:** C, C++, Scala, Python, Go, Java, TypeScript, JavaScript
- **Systems:** real-time and distributed systems, multithreading, low-latency pipelines, Linux, gRPC, Protobuf, Redis, PostgreSQL, Docker, GitLab CI/CD, Jenkins, Git